

Prediction of V_{Si}^{2-} in SiC as promising color centers for qubits

Tangjiang Qian¹, Xin-Gao Gong^{1,2} and Ji-Hui Yang^{1,2,*}

¹Key Laboratory for Computational Physical Sciences (MOE), State Key Laboratory of Surface Physics, Department of Physics, Fudan University, Shanghai 200433, China

²Hefei National Laboratory, Hefei 230088, China

(Received 26 October 2025; revised 5 March 2026; accepted 17 March 2026; published 1 April 2026)

Optically active spin defects in semiconductors and insulators, also known as color centers, have emerged as highly promising platforms for qubits. In recent years, the search for color centers that possess both excellent optical and spin properties has become a major focus of research. In this work, using first-principles calculations, we theoretically predict V_{Si}^{2-} defect in 4H- and 6H-SiC as a promising color center with high spin ($S = 1$), exhibiting a relatively weak electron-phonon coupling that leads to a large Debye-Waller factor ($>10\%$). Additionally, its NV-like electronic structure and local environment result in a significant zero-field splitting (1.16–1.29 GHz) and a long spin coherence time (>1.3 ms). Combining the advantages of known color centers, V_{Si}^{2-} in SiC exhibits outstanding optical and spin properties, which can serve as an ideal system for qubits. Furthermore, we find that V_{Si}^{2-} is also a promising single-photon source in the IR-B and IR-C region for long-distance communications, addressing the current gap in quantum-coherent materials.

DOI: [10.1103/kg61-94hg](https://doi.org/10.1103/kg61-94hg)

I. INTRODUCTION

The rapid advancement of quantum technologies promises revolutionary capabilities in computation, communication, and sensing that could surpass classical limits [1–4]. At the heart of these technologies lies the quantum bit, or qubit—the fundamental building block for quantum information processing. Various physical systems have been explored as qubit candidates, including superconducting circuits, trapped ions, photons, and semiconductor quantum dots, each with distinct advantages and limitations [5–7]. Among these, color centers in crystalline materials have emerged as highly promising qubit platforms due to their atomic-scale nature, optical control, and potential for room-temperature operation, which have attracted intensive studies in recent years [8–10]. Generally speaking, ideal color centers for qubit applications should have high spin states, large Debye-Waller factors (DWFs), significant zero-field splitting (ZFS), and long spin coherence times. High spin states enable more complex quantum operations and higher-dimensional quantum processing that can be controlled and measured with remarkable precision. Large DWFs, which indicate strong zero-phonon line emission relative to phonon sidebands, are crucial for efficient optical interfaces, quantum networking, and spin-photon entanglement protocols. Significant ZFS allows spin-state discrimination without precise external magnetic fields, facilitating room-temperature operation and resilience against magnetic noise. Most critically, long spin coherence times determine how long quantum information can be preserved, directly affecting the complexity of achievable quantum algorithms and the sensitivity of quantum sensors. The search for

and design of color centers that satisfy the above requirements has been attracting increasing attention in the field.

At present, the nitrogen vacancy (NV) center in diamond has been the most extensively studied color center due to its spin-1 ground state with about 2.87 GHz ZFS [11] and coherence times reaching milliseconds under appropriate control techniques [12,13]. However, its relatively low DWF of only 3% and emission in spectral regions with significant optical fiber attenuation limit its application in quantum networking [9,14,15]. Meanwhile, group-IV color centers (SiV^- , GeV^- , SnV^- , and PbV^-) offer improved optical properties through their crystallographic inversion symmetry (D_{3d} point group), resulting in stronger zero-phonon line emission, reduced spectral diffusion, and narrower inhomogeneous linewidth [16–20]. Yet, these centers typically require operation at cryogenic temperatures (below 0.5 K for SiV^- [21,22] and 2 K for GeV^- [23]) to suppress orbital relaxation arising from single phonon processes that limit their spin coherence times to the nanosecond regime [19,24–26]. Consequently, identifying beyond diamond-based color centers in novel platforms has begun recently, among which silicon carbide (SiC) has emerged as a particularly promising alternative host material for quantum applications [27–29]. Unlike diamond, SiC is a semiconductor with mature industrial fabrication processes, compatibility with complementary metal-oxide-semiconductor (CMOS) technology, and multiple polytypes (3C, 4H, 6H) that offer different bandgaps and crystal symmetries [30]. This diversity enables a rich variety of color centers with distinct properties not available in diamond. Color centers already identified in SiC include NV^- [31,32], VV^0 [33,34], V_{Si}^- [35], and $C_{Si}V_C^0$ [36], each offering unique advantages. For example, NV^- and VV^0 can realize coherent control at elevated temperatures, even room temperature [37–40] and have fluorescence emission around

*Contact author: jhyang04@fudan.edu.cn

the telecommunication wavelengths [41,42]. V_{Si}^- has a half-integer spin of $S = 3/2$ and a single spin orientation along the crystallographic c axis [43–45], which can permit one to remove ambiguity on the resonance transitions and be assigned without having to monitor the crystallographic orientation [46–49], thus it can be used for magnetometry applications [50,51]. $C_{Si}V_C^0$ has ZFS ranging from -2.22 to -2.16 GHz, which is larger than other color centers [36]. However, there are some limitations for the application of these color centers. For example, VV^0 has low DWF of only 5% and V_{Si}^- has small ZFS ranging from a few MHz to ~ 400 MHz [29]. Therefore, color centers that concurrently possess all four critical properties—high spin states, large DWFs, significant ZFS, and extended spin coherence times—remain elusive, representing a significant gap in the field of quantum materials. This absence of “ideal” color centers that simultaneously excel in all essential characteristics has become a fundamental bottleneck for advancing practical quantum applications beyond laboratory demonstrations.

In this work, we demonstrate through first-principles calculations that V_{Si}^{2-} defect in 4H- and 6H-SiC constitutes an outstanding color center, featuring remarkable optical and spin characteristics including high spin of $S = 1$, a large DWF exceeding 10%, a significant ZFS between 1.16 and 1.29 GHz, and a long spin coherence time surpassing 1.3 ms. By integrating the merits of established color centers, V_{Si}^{2-} in SiC presents itself as a highly favorable qubit platform. Moreover, our results also indicate that V_{Si}^{2-} exhibits promise as a single-photon emitter operating in the IR-B and IR-C region with wavelengths ranging from 1.4 to 3 μm , highlighting its potential applicability in long-distance communication systems.

II. METHODS

A. Computational details

All the calculations were performed using the Vienna *ab initio* simulation package (VASP) [52]. The Heyd-Scuseria-Ernzerhof (HSE) hybrid functional [53,54] with a mixing parameter of $\alpha = 0.28$ was used in all the calculations except for the determination of vibrational modes. The plane-wave energy cutoff for the wave-function expansion was set to 450 eV. The calculated lattice parameters are $a = 3.07$ Å, $c = 10.04$ Å for 4H-SiC and $a = 3.07$ Å, $c = 15.06$ Å for 6H-SiC. The calculated band gaps are 3.29 and 3.05 eV for 4H-SiC and 6H-SiC, respectively, which are in good agreement with the available experimental results [55,56]. A $6 \times 6 \times 2$ supercell including 576 atomic sites was used for 4H-SiC and $6 \times 6 \times 1$ supercell including 432 atomic sites was used for 6H-SiC to model defects which is sufficient to minimize the periodic defect-defect interaction. For the Brillouin zone integration, the single Γ -point scheme was adopted. During structural relaxations, the conjugate gradient algorithm was used until the force on each atom was below 0.01 eV Å^{-1} and the total energy was converged within $1.0 \times 10^{-5} \text{ eV}$. Spin polarization was considered for the calculations of all the defects. The excited states were calculated by ΔSCF (self-consistent field) method [57]. To solve the problems with the

Jahn-Teller distortion in the excited state, we used fractional occupation numbers for degenerate states.

For the vibrational modes, we calculated the corresponding dynamical matrix containing the second-order derivatives of the total energy by means of the Perdew-Burke-Ernzerhof (PBE) functional [58] with the phonopy package [59]. In this procedure, we apply strict threshold parameters in the geometry optimization for the convergence of the electronic structure (10^{-6} eV) and atomic forces ($10^{-3} \text{ eV Å}^{-1}$). These vibrational modes were used together with the HSE ground state and excited state geometries for the calculation of the photoluminescence (PL) spectrum.

B. Optical properties

For the calculation of the PL spectrum, we used Franck-Condon approximation [60]. The phonon sideband of the fluorescence spectrum is given by [61]

$$I(\hbar\omega) = \frac{n_D \omega^3}{3\epsilon_0 \pi c^3 \hbar} |\vec{\mu}_{eg}|^2 \sum_m |\langle \chi_{gm} | \chi_{e0} \rangle|^2 \delta(E_{ZPL} - E_{gm} - \hbar\omega), \quad (1)$$

where ω is the photon frequency, n_D is the refractive index of the material, and $\vec{\mu}_{eg}$ is the electronic transition dipole moment between excited and ground states. χ_{gm} and χ_{e0} are the vibrational levels of the ground and excited states, respectively. E_{ZPL} is the energy difference between excited and ground states, E_{gm} is energy of the ground state.

To compare with experimental spectra, we used a normalized luminescence intensity:

$$L(\hbar\omega) = CA(\hbar\omega), \quad (2)$$

where

$$A(\hbar\omega) = \sum_m |\langle \chi_{gm} | \chi_{e0} \rangle|^2 \delta(E_{ZPL} - E_{gm} - \hbar\omega) \quad (3)$$

and the normalization constant C is given by

$$C^{-1} = \int A(\hbar\omega) \omega^3 d(\hbar\omega). \quad (4)$$

$A(\hbar\omega)$ is obtained from the Fourier transform of the generating function $G(t)$ [62,63]:

$$A(E_{ZPL} - \hbar\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} G(t) e^{i\omega t - \gamma|t|} dt, \quad (5)$$

where γ represents the broadening of the zero-phonon line (ZPL). $G(t)$ is defined as

$$G(t) = e^{S(t) - S(0)}, \quad (6)$$

where $S(t)$ is defined as

$$S(t) = \int_0^\infty S(\hbar\omega) e^{-i\omega t} d(\hbar\omega). \quad (7)$$

The total Huang-Rhys factor (HRF), representing the average number of phonons involved in the emission, is obtained as

$$S = S(0) = \int_0^\infty S(\hbar\omega) d(\hbar\omega) = \sum_k S_k, \quad (8)$$

where $S(\hbar\omega)$ is the spectral function of electron-phonon coupling, and S_k is the partial HRF for the phonon mode k , which is defined as

$$S_k = \frac{\omega_k q_k^2}{2\hbar} \quad (9)$$

with

$$q_k = \sum_{\alpha i} m_\alpha^{1/2} (R_{e;\alpha i} - R_{g;\alpha i}) \Delta r_{k;\alpha i}. \quad (10)$$

Here, α denotes atoms, i represents direction (x, y, z), m is the mass of atom α , $R_{e;\alpha i}$ and $R_{g;\alpha i}$ are the equilibrium positions in the excited and the ground states, respectively. $\Delta r_{k;\alpha i}$ is the normalized vector which represents the displacement of the atom α along the direction i in the phonon mode k .

The mass-weighted structural displacement Q is calculated as

$$Q = \sqrt{\sum_{\alpha i} m_\alpha (R_{e;\alpha i} - R_{g;\alpha i})^2}, \quad (11)$$

which represents the difference between the geometries of the ground and excited states, and the DWF is defined as below:

$$DWF = e^{-S}, \quad (12)$$

which can be estimated from the ratio of intensity between the ZPL and the total PL emissions.

The radiative lifetime of the zero-phonon line (ZPL) transition is calculated from the Einstein coefficient A as

$$\tau_{\text{rad}} = \frac{3\pi\epsilon_0\hbar c^3}{n_D\omega^3 |\bar{\mu}_{eg}|^2}, \quad (13)$$

where, ϵ_0 , c , ω are the vacuum permittivity, the speed of light, the transition frequency of ZPL , respectively. n_D is the refractive index. $\bar{\mu}_{eg}$ can be calculated by

$$\bar{\mu}_{eg} = \frac{i\hbar}{(\epsilon_f - \epsilon_i)m} \langle \psi_f | \vec{p} | \psi_i \rangle, \quad (14)$$

where ψ_i and ψ_f are the initial orbital from the wave function of the ground state and final orbital from the wave function of the excited state, respectively. ϵ_i and ϵ_f are their eigenvalues and m is the electron mass.

C. Spin properties

The ZFS parameters were calculated by employing the vaspprojector-augmented-wave implementation of electron spin-spin interaction [11,64]. The ZFS tensor D_{ab} can be written as

$$D_{ab} = \frac{1}{2S(2S-1)} \frac{\mu_0 g_e^2 \mu_B^2}{4\pi} \sum_{m < n} \chi_{mn} \int |\Psi_{mn}(\vec{r}_1, \vec{r}_2)|^2 \times \frac{r^2 \delta_{ab} - 3r_a r_b}{r^5} d\vec{r}_1 d\vec{r}_2, \quad (15)$$

where S , μ_0 , g_e , μ_B are the total spin, the vacuum magnetic permeability, the Landé g factor of free electrons, and the Bohr magneton, respectively. χ_{mn} is $+1(-1)$ for the parallel (antiparallel) spins. $\Psi_{mn}(\vec{r}_1, \vec{r}_2)$ is the Slater determinant of the Kohn-Sham orbitals ψ_m and ψ_n . r , r_a , and r_b are the absolute

value and the components of the vector $\vec{r}_1 - \vec{r}_2$, respectively. After diagonalization, the ZFS tensor is a second-order trace-less tensor, denoted as ${}^dD(D_{xx}, D_{yy}, D_{zz})$. The axial and transversal components of the ZFS parameters D and E are defined as

$$D = \frac{3}{2} D_{zz}, \quad E = \frac{D_{yy} - D_{xx}}{2}. \quad (16)$$

Here we also consider the problem of spin contamination. By eliminating the spin contamination of the two-particle spin density [65], we calculated the triplet dD_t and the averaged singlet dD_s . The final corrected ZFS tensor ${}^d\tilde{D}$ will be brought by

$${}^d\tilde{D} = \frac{{}^dD_t - {}^dD_s}{2} \quad (17)$$

and the corrected ZFS parameters D and E will finally be given using Eq. (14) with ${}^d\tilde{D}$ replacing dD .

The hyperfine coupling constants were calculated following the method of Ref. [66]. The hyperfine tensor A_{ab} which describes the interaction between the electron spin and nuclear spin (located at site $\vec{R}$) is given as

$$A_{ab}^{(n)} = \frac{2\mu_0}{3} g_e \mu_B g_n \mu_n \frac{\rho_S(\vec{R})}{S} + \frac{\mu_0}{4\pi} g_e \mu_B g_n \mu_n \frac{1}{S} \int \frac{3r_a r_b - r^2 \delta_{ab}}{r^5} \rho_S(\vec{r}) d^3\vec{r}, \quad (18)$$

where $\vec{r}$ denotes the direction vector between the electron and nuclear spin. r , r_a and r_b are the absolute value and the components of the vector $\vec{r}$, respectively. g_n is the g factor of the nucleus, while μ_n is the nuclear magneton. $\rho_S(\vec{r})$ refers to the spin density of the electron as a function of $\vec{r}$. $\rho_S(\vec{R})$ refers to the spin density located at site $\vec{R}$. The first and second terms represent the Fermi contact and the magnetic dipole-dipole interaction terms, respectively. We used the HSE functional to calculate the hyperfine tensor. The core electronic contributions to the Fermi contact term are calculated in the frozen valence approximation [67].

We calculated electron spin coherence time T_2 of the defect using the generalized cluster-correlation expansion approach implemented in the PyCCE [68]. The coherence function $\mathcal{L}(t)$ that describes the loss of relative phase of qubit states can be calculated as below:

$$\mathcal{L}(t) = \frac{\langle 1 | \rho_S(t) | 0 \rangle}{\langle 1 | \rho_S(0) | 0 \rangle} = \frac{\langle \hat{\sigma}_-(t) \rangle}{\langle \hat{\sigma}_-(0) \rangle}, \quad (19)$$

where $\rho_S(t)$ is the density matrix of the central spin. $|0\rangle$ and $|1\rangle$ represent qubit levels. We can obtain T_2 through an exponential fitting towards the envelope of $\mathcal{L}(t)$ [69]:

$$\mathcal{L}(t) = e^{-(\frac{t}{T_2})^n}. \quad (20)$$

In this work, we calculated hyperfine values using the methodology described above, incorporating nuclear spins located up to 10 Å from the defect center. For more distant nuclear spins, only the direct dipole-dipole interaction was considered. We employed a supercell with a linear dimension of 200 Å and incorporated natural isotopic abundances (^{13}C : 1.11%, ^{29}Si : 4.70%) to accurately represent realistic systems. To ensure computational convergence, we established boundary conditions limiting the maximum distance

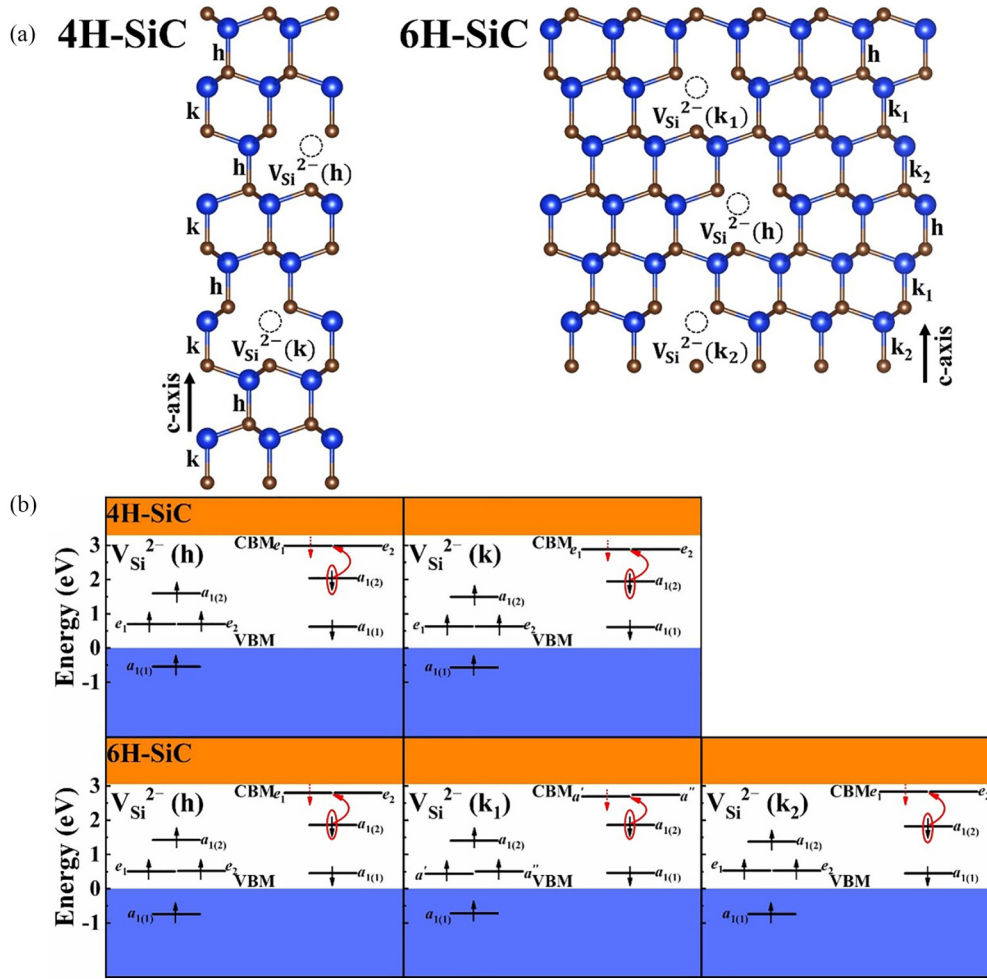


FIG. 1. The atomic configurations and electronic structures of V_{Si}^{2-} . (a) The atomic configurations of V_{Si}^{2-} in 4H-SiC (left panel) and 6H-SiC (right panel). Blue atoms represent Si, brown atoms represent C, and the dashed circles indicate the Si vacancy. (b) The single-particle defect levels of V_{Si}^{2-} in 4H-SiC (upper panel) and 6H-SiC (lower panel), referred to the VBM. The upward and downward arrows stand for the up- and down-spin states of the occupied electrons, respectively. The red dashed arrows represent the electrons excited from the circled defect levels.

between the central spin and bath spins to 80 Å, while restricting the maximum separation between bath spins to 10 Å. All reported results represent ensemble averages across 200 distinct nuclear spin bath configurations, providing statistically robust hyperfine interaction.

III. RESULTS

A. Atomic configurations and electronic structures of V_{Si}^{2-}

First, we discuss the configurations and electronic structures of V_{Si}^{2-} . We note that previous studies indicate that V_{Si}^{2-} in 3C-SiC is not stable against decomposing into 0 and -1 states and the defect level is highly delocalized [70,71]. In contrast, V_{Si}^{2-} in 4H- and 6H-SiC can stably exist in n-type doped samples when the Fermi level is close to the conduction band minimum (CBM) [72–74]. Previous studies [32,75] have shown that the charge transition levels (0/–) and (–/2–) for both V_{Si} and $N_C V_{Si}$ in 4H- and 6H-SiC are located at approximately 1.5 and 2.6 eV above the valence band maximum (VBM), respectively. This implies that the V_{Si}^- and NV^- color centers are thermodynamically stable

when the Fermi level lies between 1.5 and 2.6 eV above the VBM. In contrast, due to the (2–/3–) charge transition level located around 2.9 eV above the VBM, V_{Si}^{2-} is most stable when the Fermi level is positioned between 2.6 and 2.9 eV above the VBM. Under these conditions, the formation energy of V_{Si}^{2-} is estimated to range from 5.9 to 6.5 eV, which is lower than that of V_{Si}^- under typical material synthesis conditions, confirming the thermodynamic favorability of V_{Si}^{2-} . Therefore, in the following, we focus on studying color center properties of V_{Si}^{2-} in 4H- and 6H-SiC. As shown in Fig. 1(a), V_{Si}^{2-} in 4H-SiC has two inequivalent configurations, labeled as $V_{Si}^{2-}(h)$ and $V_{Si}^{2-}(k)$, whereas in 6H-SiC, it exhibits three configurations: $V_{Si}^{2-}(h)$, $V_{Si}^{2-}(k_1)$ and $V_{Si}^{2-}(k_2)$ [see Fig. 1(a)]. All of these configurations exhibit C_{3v} point group symmetry, except for $V_{Si}^{2-}(k_1)$, which has a reduced symmetry of C_{1h} after the structural relaxation due to the Jahn-Teller distortion. The defect levels of V_{Si}^{2-} are shown in Fig. 1(b). In general, V_{Si}^{2-} with different configurations exhibit similar single-particle defect levels, characterized by two nondegenerate states [$a_{1(1)}$ and $a_{1(2)}$] and a doubly degenerate e state. For $V_{Si}^{2-}(k_1)$, due to its lower symmetry, the degenerate e

states are slightly split with only 65.8 and 45.7 meV for the spin-up and spin-down channels, respectively. Notably, the defect levels of V_{Si}^{2-} differ significantly from those of V_{Si}^- , which exhibits near T_d symmetry, resulting in nearly degenerate $a_{1(2)}$ and e states [76].

In total, V_{Si}^{2-} creates eight defect levels with six electrons occupying the lower states, leaving the two spin-down e levels empty. Except that the $a_{1(1)}$ level in the spin-up channel lies below the VBM, all the other seven levels lie within the band gap, which originate from the four dangling bonds of C atoms around the Si vacancy with the $a_{1(2)}$ defect level mainly localized on the C atom on the symmetry axis of the isolated silicon vacancy and the e level mainly localized on the remaining three C atoms out of the symmetry axis of the isolated silicon vacancy. It is worth noting that in the spin-up channel, the $a_{1(2)}$ level lies approximately 0.9 eV above the e level, which is the opposite of the ordering observed in NV^- , VV^0 , and V_{Si}^- [77,78]. In the spin-down channel, the electron at the $a_{1(2)}$ level can be excited to the e levels, indicating key color center properties. In addition, the six defect electrons associated with V_{Si}^{2-} contribute to a total spin of $S = 1$, making V_{Si}^{2-} a high-spin color center with an electronic structure similar to NV center in SiC.

However, we note that the color center properties of V_{Si}^{2-} are rarely reported experimentally. This can be understood as follows. The $a_{1(2)}$ levels in the spin-down channel are approximately 1.25 and 1.34 eV below the CBM of 4H-SiC, for $V_{Si}^{2-(h)}$ and $V_{Si}^{2-(k)}$, respectively [see Fig. 1(b)]. For 6H-SiC, the $a_{1(2)}$ levels in the spin-down channel are located approximately 1.19, 1.19, and 1.23 eV below the CBM for $V_{Si}^{2-(h)}$, $V_{Si}^{2-(k_1)}$ and $V_{Si}^{2-(k_2)}$, respectively [see Fig. 1(b)]. Experimentally, the photoexcitation energies for probing V_{Si}^- in SiC are usually larger than 1.3 eV, as it is well known that V_{Si}^- color centers have ZPL ranging from 1.35 to 1.44 eV [35,73]. Consequently, the electron in the spin-down $a_{1(2)}$ level of V_{Si}^{2-} can be directly excited into the conduction band, turning V_{Si}^{2-} to V_{Si}^- . Therefore, it is not possible to detect color center properties of V_{Si}^{2-} unless lower photoexcitation energies, i.e., around 0.7 eV, are used.

B. Optical properties

Next, we discuss the optical properties of V_{Si}^{2-} color centers. Figure 2 presents the configuration coordinate diagram illustrating the definitions of the optical parameters; i.e., the absorption, emission and ZPL energies. For V_{Si}^{2-} , the excited state corresponds to the state with the e level occupied and the $a_{1(2)}$ level unoccupied in the spin-down channel, as discussed above. It should be noted that for the $V_{Si}^{2-(k_1)}$ in 6H-SiC, the order of the occupied e level and the empty a level in the excited state are interchanged after structural relaxation (as shown in Fig. S1 of the Supplemental Material [79]), indicating that $V_{Si}^{2-(k_1)}$ in 6H-SiC does not form a stable emitter; therefore, we will not discuss the optical properties of $V_{Si}^{2-(k_1)}$ further. For other V_{Si}^{2-} configurations in 4H- and 6H-SiC, they are all stable emission centers. The calculated absorption, emission and ZPL energies are summarized in Table I.

As shown in Table I, $\Delta\epsilon$ is significantly larger than the corresponding absorption energy, which can be attributed

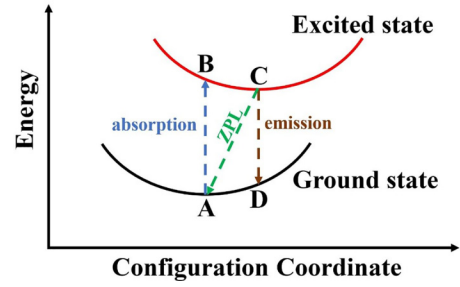


FIG. 2. Configuration coordinate diagram for the ground and excited states of a color center. Electron capture processes including phonon-assisted absorption (blue arrow), phonon-assisted emission (brown arrow) and zero-phonon line (ZPL) optical transition (green arrow) are shown in the diagram.

to electron-hole coupling. This coupling shifts the defect levels, resulting in the difference between $\Delta\epsilon$ and the absorption energy. In addition, the absorption and emission energies differ noticeably from those of ZPLs due to phonon involvement during the optical transition. Nevertheless, the energy ordering remains unchanged in either case, i.e., $V_{Si}^{2-(k)} > V_{Si}^{2-(h)}$ in 4H-SiC and $V_{Si}^{2-(k_2)} > V_{Si}^{2-(h)}$ in 6H-SiC. The calculated ZPL energies of $V_{Si}^{2-(h)}$ in 4H- and 6H-SiC are 0.524 and 0.520 eV, respectively, while the ZPLs of $V_{Si}^{2-(k)}$ in 4H- and 6H-SiC are 0.555 and 0.614 eV, respectively. It should be noted that we used fractional occupation numbers $a_1(1)e_x(1.5)e_y(1.5)$ for degenerate states, thus neglecting Jahn-Teller (JT) distortions, which is a widely adopted and well-justified practical strategy [63,80–82]. Specifically, to circumvent the problems associated with the JT distortion arising from the degeneracy of the $a_1(1)e_x(2)e_y(1)$ and $a_1(1)e_x(1)e_y(2)$ configurations, the excited-state charge density was constrained to be the average of the two degenerate configurations, retaining the C_{3v} symmetry. We also compared the results obtained with fractional and integer occupation numbers (see Table S1 in the Supplemental Material [79]) to examine the JT reconstruction of the ZPLs. Overall, we found a systematic decrease of approximately 0.15–0.17 eV for all configurations of V_{Si}^{2-} when the JT effect is explicitly taken into account without symmetry constraints. This systematic shift does not affect the ordering of the ZPLs of V_{Si}^{2-} , and our primary conclusions remain unchanged. Notably, after considering JT distortions, all the ZPLs of V_{Si}^{2-} which are relevant for quantum communication

TABLE I. Calculated values of $\Delta\epsilon$, ZPL energy, absorption energy and emission energy for the V_{Si}^{2-} defect in 4H and 6H-SiC. $\Delta\epsilon$ denotes the energy difference between spin-down $a_{1(2)}$ and e levels in the ground states of V_{Si}^{2-} (unit: eV).

Configuration	$\Delta\epsilon$	ZPL	Absorption	Emission
4H – SiC				
$V_{Si}^{2-(h)}$	0.947	0.524	0.825	0.360
$V_{Si}^{2-(k)}$	0.935	0.555	0.841	0.396
6H – SiC				
$V_{Si}^{2-(h)}$	0.936	0.520	0.811	0.371
$V_{Si}^{2-(k_2)}$	1.010	0.614	0.896	0.470

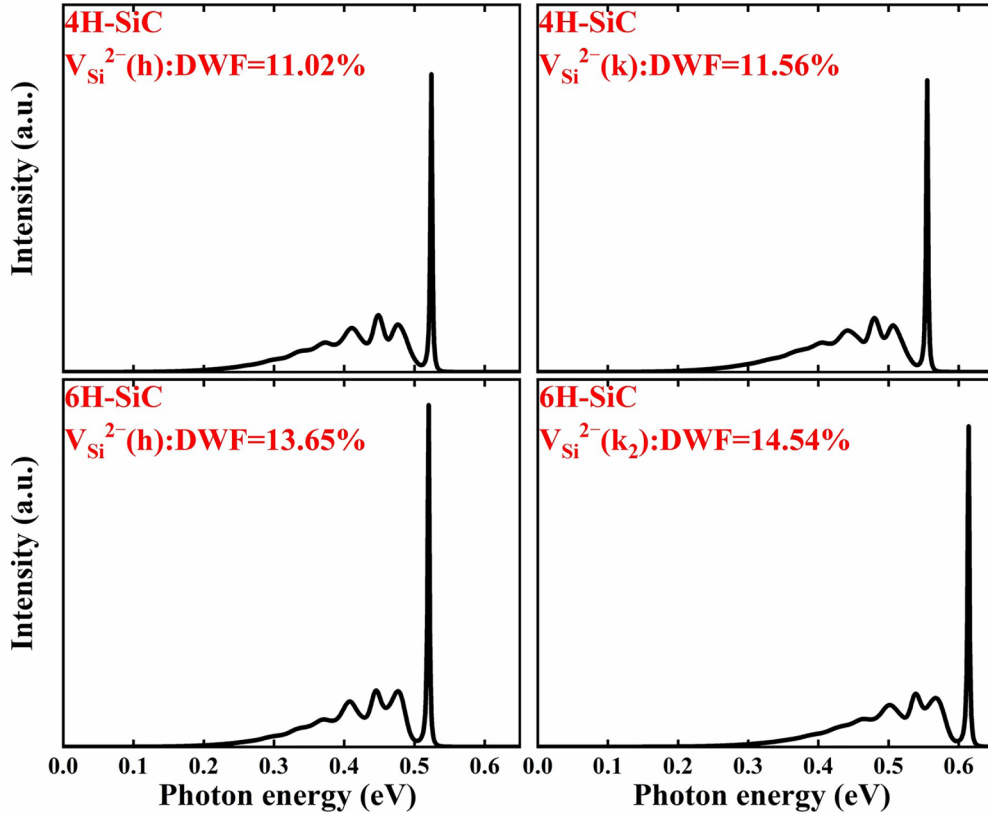


FIG. 3. Calculated normalized photoluminescence line shapes $L(\hbar\omega)$ for V_{Si}^{2-} in 4H- (upper panel) and 6H-SiC (lower panel). It should be noted that we use the normalized luminescence spectra $L(\hbar\omega)$ to compare the PL line shapes among different defects. However, the PL intensities of different defects cannot be directly compared, since their absolute intensities are quite different.

applications lie in the range of 0.36–0.46 eV, corresponding to the IR-B (1.4–3 μm) and IR-C (3 μm –1 mm) region [83]. This is significant, as there is a recognized scarcity of single-photon sources in this spectral range among quantum-coherent materials for long-distance communications [8,84], a limitation that hinders applications in quantum networks and integrated photonic circuits. Therefore, V_{Si}^{2-} might represent a promising candidate to address this pressing need.

Furthermore, we calculate the PL spectra for V_{Si}^{2-} with different configurations. As shown in Fig. 3, the PL spectra of V_{Si}^{2-} in both 4H- and 6H-SiC are nearly identical, all featuring a broadening of approximately 0.3 eV due to their shared C_{3v} symmetry. The subtle differences arise from the distinct local environments at the h and k sites in 4H-SiC, and the h and k_2 sites in 6H-SiC. These distinctions are quantitatively reflected in the computed ΔQ values, which are 0.557/0.552 $\text{\AA emu}^{1/2}$ for the h/k sites in 4H-SiC and 0.534/0.521 $\text{\AA emu}^{1/2}$ for the h/ k_2 sites in 6H-SiC. The corresponding total HRFs are 2.205/2.158 for 4H-SiC and 1.992/1.928 for 6H-SiC. The calculated DWFs are notably high: 11.02% and 11.56% for the h- and k-site configurations in 4H-SiC, and even larger values of 13.65% and 14.54% for the h- and k_2 -site configurations in 6H-SiC. Remarkably, these DWF values significantly exceed those of several well-studied color centers, such as NV^- center in diamond (3.2%) [63], VV^0 center in SiC (5%) [85] and V_{Si}^- center in SiC (6%) [35]. To further elucidate the origin of this enhancement, we

compare the PL spectra and spectral functions of $V_{\text{Si}}^{2-}(\text{h})$ and $V_{\text{Si}}^{2-}(\text{h})$ in 4H-SiC as shown in Fig. S3 (see Supplemental Material [79] for details). The results indicate that $V_{\text{Si}}^{2-}(\text{h})$ exhibits substantially weaker electron-phonon coupling in the low-frequency region, leading to a smaller total HRF and consequently a DWF nearly double that of $V_{\text{Si}}^{2-}(\text{h})$. This high DWF signifies a dominant ZPL contribution in the emission spectrum, underscoring the potential of V_{Si}^{2-} as a high-performance infrared single-photon source operating in the IR-B and IR-C region. We also note that the DWF of $V_{\text{Si}}^{2-}(\text{k})$ in 4H-SiC was calculated to be 22.97% by Hashemi *et al.* [81], roughly twice the value obtained in the present work. This discrepancy can be attributed to the differences in computational methodology: in Ref. [81], the structures were first optimized using the PBEsol functional, then the lattice parameters were rescaled to match the HSE06 results, followed by one-shot HSE06 total-energy calculations without further structural relaxations. In contrast, in the present study both the ground-state and excited-state geometries are fully optimized at the HSE level, ensuring full internal consistency between the exchange-correlation functional and the atomic structures—a crucial factor for accurately describing defect-related properties. Moreover, we employ a larger 576-atom supercell, compared to the 400-atom supercell used in Ref. [81], which further reduces periodic defect-defect interactions. To substantiate the quantitative reliability of our results, we performed a convergence test by computing the

TABLE II. Calculated ZFS parameters of V_{Si}^{2-} defects in 4H- and 6H-SiC with HSE functional (after spin decontamination), which involves axial component ($\tilde{D}$) and transversal component ($\tilde{E}$) from zero-field splitting tensor.

Configuration	$\tilde{D}; \tilde{E}$ (MHz)
4H – SiC	
$V_{Si}^{2-}(h)$	1253.35;0.24
$V_{Si}^{2-}(k)$	1227.19;0.02
6H – SiC	
$V_{Si}^{2-}(h)$	1217.89;4.11
$V_{Si}^{2-}(k_1)$	1163.61;58.73
$V_{Si}^{2-}(k_2)$	1291.20;1.37

DWF of $V_{Si}^{2-}(h)$ in 4H-SiC using both 400-atom and 576-atom supercells and found that the difference is within 10% (see Fig. S2 in the Supplemental Material [79]). These tests demonstrate that our calculated results are robust and well converged. Accordingly, the present values are expected to provide a more reliable reference and extend the earlier study to other configurations of V_{Si}^{2-} in SiC.

Last but not least, we calculate the radiative transition lifetime (τ_{rad}) connecting the JT distorted excited state to the high-symmetry ground state for V_{Si}^{2-} , as shown in Table S2 in the Supplemental Material [79]. The calculated radiative lifetimes are in the range of 55–122 ns, which makes it possible to observe V_{Si}^{2-} at a single defect level. In addition, we have calculated the spin-conserving nonradiative transition lifetime (τ_{IC}) of $V_{Si}^{2-}(h)$ in 4H-SiC using the Nonrad code [86]. The calculated τ_{IC} at 300 K is 1.26 μ s, which is significantly longer than the radiative transition lifetime ($\tau_{rad} = 110$ ns); this indicates that the internal conversion rate should be much smaller than the radiative transition rate. Taken together, our results support it as a promising candidate for a solid-state quantum emitter.

C. Spin properties

Finally, we discuss the spin characteristics of V_{Si}^{2-} . To gain insights into its spin properties, we examine the local environment of V_{Si}^{2-} in Fig. S4 and Fig. S5 of the Supplemental Material [79]. As shown in Fig. S4, the spin density of $V_{Si}^{2-}(S = 1)$ is mainly localized on the three C atoms out of the symmetry axis of the isolated silicon vacancy, which is similar to that of NV^- and VV^0 [87], suggesting that they may share similar spin properties. Furthermore, we analyze the local structure of V_{Si}^{2-} for all the configurations in 4H- and 6H-SiC as shown in Fig. S5. We find that the C-C distances between the three C atoms out of the symmetry axis of the isolated silicon vacancy range from 3.31 to 3.35 Å, while the Si-C bond length corresponding to the C atom on the symmetry axis of the isolated silicon vacancy is around 1.78 Å. These values are very close to those of NV centers in SiC [80], indicating a highly similar local structure of V_{Si}^{2-} . Taken together, we can conclude that V_{Si}^{2-} has an NV-like electronic structure and local environment, indicating that it exhibits similar spin properties to the NV center in SiC. The calculated ZFS parameters of the V_{Si}^{2-} defects are summarized in Table II. Prior to spin decontamination, the calculated values

of ZFS parameter $D(E)$ are 2002(0.48) and 1909(0.04) MHz for $V_{Si}^{2-}(h)$ and $V_{Si}^{2-}(k)$ in 4H-SiC, and 1982(8), 1891(117) and 2003(3) MHz for $V_{Si}^{2-}(h)$, $V_{Si}^{2-}(k_1)$, and $V_{Si}^{2-}(k_2)$ in 6H-SiC, respectively. After spin decontamination, the corrected $\tilde{D}(\tilde{E})$ are 1253(0.24) and 1227(0.02) MHz for $V_{Si}^{2-}(h)$ and $V_{Si}^{2-}(k)$ in 4H-SiC, and 1218(4), 1164(59) and 1291(1) MHz for $V_{Si}^{2-}(h)$, $V_{Si}^{2-}(k_1)$, and $V_{Si}^{2-}(k_2)$ in 6H-SiC, respectively. The E component is negligible compared to the D component for $V_{Si}^{2-}(h)$ and $V_{Si}^{2-}(k)$ in 4H-SiC, as well as for $V_{Si}^{2-}(h)$ and $V_{Si}^{2-}(k_2)$ in 6H-SiC, which is consistent with their C_{3v} symmetry. In contrast, the $V_{Si}^{2-}(k_1)$ in 6H-SiC exhibits a significant E component due to its lower C_{1h} symmetry. Physically, the D parameter quantifies the energy separation between the $m_S = \pm 1$ sublevels and the $m_S = 0$ sublevel, while the E value reflects the subsequent splitting of $m_S = \pm 1$ sublevels in the absence of external magnetic field. The D values of V_{Si}^{2-} lie within the range 1.16–1.29 GHz, which is comparable to those of NV^- and VV^0 in SiC [29,88]. Moreover, we observe the variation in D values among different lattice sites, which provides a spectroscopic fingerprint to distinguish the local configurations of V_{Si}^{2-} in 4H- and 6H-SiC. The order of D values is $V_{Si}^{2-}(h) > V_{Si}^{2-}(k)$ in 4H-SiC, and $V_{Si}^{2-}(k_2) > V_{Si}^{2-}(h) > V_{Si}^{2-}(k_1)$ in 6H-SiC. This variation can be explained by the different C-C distances corresponding to the three C atoms out of the symmetry axis of the isolated silicon vacancy among these configurations (as shown in Fig. S5 of the Supplemental Material [79]), which indicates that the order of C-C distances is $V_{Si}^{2-}(h) < V_{Si}^{2-}(k)$ in 4H-SiC, and $V_{Si}^{2-}(k_2) < V_{Si}^{2-}(h) < V_{Si}^{2-}(k_1)$ in 6H-SiC. Since the ZFS is mainly determined by the coupling between the e levels in the spin-up channel, which correspond to the $S = 1$ ground state and are primarily localized on the three C atoms out of the symmetry axis of the isolated silicon vacancy, the D values of V_{Si}^{2-} are expected to be inversely related to the C-C distances. As a result, the difference in C-C distance accounts for the deviation in D values across different configurations. It should be noted that the present work focuses on the ZFS in the ground state of V_{Si}^{2-} at $T = 0$ K, which is not JT active. However, the ZFS in the polaronic excited state and its temperature dependence are also essential in understanding the intersystem-crossing processes [89], which goes beyond the scope of the current paper.

We also calculated the principal values of the hyperfine tensor for the V_{Si}^{2-} defects, which can be used to illustrate the coupling strength of the defect spins with nearby nuclear spins and identify defects. As shown in Fig. 4, V_{Si}^{2-} has four first-neighbor C atoms and 12 second-neighbor Si atoms. The four first-neighbor C atoms are divided into C_{1a} and C_{1b} according to their geometrical positions. Each of these C atoms is bonded to three second-neighbor Si atoms. As shown in Table III, we compute the principal values of the hyperfine tensor for the first- and second-neighbor atoms. We consider the ^{13}C and ^{29}Si isotopes, both of which have nuclear spin of $I = 1/2$. The $^{13}C_{1a}$ atoms exhibit principal values ranging from 3 to 5 MHz, while the $^{13}C_{1b}$ atoms show significantly higher values, ranging from 30 to 117 MHz. This difference arises because the spin density is mainly located on the three $^{13}C_{1b}$ atoms, in agreement with the illustrated spin density in Fig. 4. This spin density distribution differs significantly from that

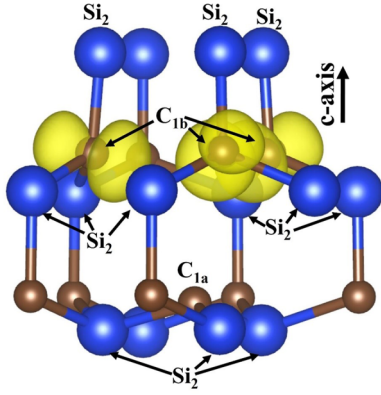


FIG. 4. The atomic environment of $V_{\text{Si}}^{2-}(\text{h})$ in 4H-SiC (viewed perpendicular to the c axis) showing the nearest C atoms and second nearest Si atoms (Other V_{Si}^{2-} in 4H- and 6H-SiC exhibit similar atomic environments). Yellow lobes show the spin density distributions with the isosurface set as 10% of the maximum amplitude. The sites are numbered according to their distance from the silicon vacancy. The atom labels indicate the neighboring nuclei around the isolated silicon vacancy: the single on-axis first-neighbor ^{13}C (C_{1a}), the three equivalent off-axis first-neighbor ^{13}C atoms (C_{1b}), and the 12 second-neighbor ^{29}Si sites (Si_2).

of the V_{Si}^{2-} color center [76]. For the 12 $^{29}\text{Si}_2$ atoms, which are connected to the four ^{13}C atoms with dangling bonds, the principal values lie around 8–11 MHz, slightly larger than those of $^{13}\text{C}_{1a}$, consistent with the spin density distribution

TABLE III. The calculated total hyperfine coupling parameters for V_{Si}^{2-} defects in 4H and 6H-SiC with the HSE functional. The hyperfine splittings are determined for the first-neighbor ^{13}C nuclei (C_{1a} and C_{1b}) and the 12 second-neighbor ^{29}Si sites (Si_2). $^{13}\text{C}_{1a} \times 1$ refers to the first-neighbor ^{13}C nuclei on the symmetry axis of the isolated silicon vacancy; $^{13}\text{C}_{1b} \times 3$ indicates the three equivalent first-neighbor ^{13}C nuclei out of the symmetry axis of the isolated silicon vacancy; $^{29}\text{Si}_2 \times 12$ refers to the 12 second-neighbor ^{29}Si sites with equivalent positions.

Configuration	Sites	A_{xx} (MHz)	A_{yy} (MHz)	A_{zz} (MHz)
4H – SiC				
$V_{\text{Si}}^{2-}(\text{h})$	$^{13}\text{C}_{1a} \times 1$	−4.429	−4.130	−4.429
	$^{13}\text{C}_{1b} \times 3$	43.996	43.294	115.115
	$^{29}\text{Si}_2 \times 12$	8.747	8.354	9.227
$V_{\text{Si}}^{2-}(\text{k})$	$^{13}\text{C}_{1a} \times 1$	−4.351	−3.816	−4.351
	$^{13}\text{C}_{1b} \times 3$	34.585	33.904	105.426
	$^{29}\text{Si}_2 \times 12$	9.264	8.937	9.635
6H – SiC				
$V_{\text{Si}}^{2-}(\text{h})$	$^{13}\text{C}_{1a} \times 1$	−4.471	−3.580	−4.471
	$^{13}\text{C}_{1b} \times 3$	43.514	42.831	114.758
	$^{29}\text{Si}_2 \times 12$	8.803	8.211	9.160
$V_{\text{Si}}^{2-}(\text{k}_1)$	$^{13}\text{C}_{1a} \times 1$	−4.125	−3.038	−4.175
	$^{13}\text{C}_{1b} \times 3$	33.226	32.586	104.360
	$^{29}\text{Si}_2 \times 12$	10.009	9.699	10.395
$V_{\text{Si}}^{2-}(\text{k}_2)$	$^{13}\text{C}_{1a} \times 1$	−4.917	−4.320	−4.917
	$^{13}\text{C}_{1b} \times 3$	45.275	44.553	116.586
	$^{29}\text{Si}_2 \times 12$	8.650	8.214	9.074

TABLE IV. Calculated coherence time (T_2) of V_{Si}^{2-} defects in 4H- and 6H-SiC, computed at a magnetic field of 500 G.

Configuration	T_2 (ms)
4H – SiC	
$V_{\text{Si}}^{2-}(\text{h})$	1.3476
$V_{\text{Si}}^{2-}(\text{k})$	1.3818
6H – SiC	
$V_{\text{Si}}^{2-}(\text{h})$	1.3603
$V_{\text{Si}}^{2-}(\text{k}_1)$	1.4234
$V_{\text{Si}}^{2-}(\text{k}_2)$	1.4998

around the Si vacancy. Notably, the calculated total hyperfine coupling parameters for V_{Si}^{2-} are quantitatively comparable to those reported from similar *ab initio* considerations for the NV centers in SiC [32], supporting the NV-like local environment of V_{Si}^{2-} . Regarding the four first-neighbor carbon atoms, in 4H-SiC, $V_{\text{Si}}^{2-}(\text{h})$ exhibits slightly larger hyperfine values than the $V_{\text{Si}}^{2-}(\text{k})$. In 6H-SiC, the order follows: $V_{\text{Si}}^{2-}(\text{k}_2) > V_{\text{Si}}^{2-}(\text{h}) > V_{\text{Si}}^{2-}(\text{k}_1)$. These differences in hyperfine interactions can serve as fingerprints to distinguish different configurations of V_{Si}^{2-} .

Last but not least, we analyze the coherence time of V_{Si}^{2-} under the external magnetic field of 500 G, as shown in Fig. 5 and Table IV. Since hyperfine coupling to lattice spins is the dominant source of spin decoherence at natural abundance in SiC, we can expect long coherence times of V_{Si}^{2-} due to its hyperfine coupling parameters being similar to those of the divacancy and NV centers. Figure 5 visualizes the calculated $\mathcal{L}(t)$ and their fitting curves, and the extracted T_2 values are summarized in Table IV. The T_2 values of V_{Si}^{2-} range from 1.34 to 1.50 ms, comparable to those of other promising qubits [80,90,91], indicating that V_{Si}^{2-} exhibits excellent coherence performance for quantum technology and can be used as a potential qubit. It should be noted that the T_2 value of V_{Si}^{2-} in 6H-SiC is slightly higher than that in 4H-SiC, which can be attributed to the larger E value of V_{Si}^{2-} in 6H-SiC, providing enhanced coherence protection against external magnetic noise [92].

IV. DISCUSSION

A. Optical spin-polarization loop

We have systematically investigated the optical and spin properties of V_{Si}^{2-} in SiC, indicating that it can serve as a promising candidate for qubits. However, the optical spin-polarization loop has not been examined in this work. The key feature enabling such loops in solid-state qubits and emitters is the presence of spin-dependent processes during excitation and relaxation, which allow optical cycles to selectively polarize the defect's spin state [93]. Accurately describing this loop requires precise calculation of nonradiative intersystem crossing from the excited triplet state to an excited singlet state, involving spin-orbit and electron-phonon couplings between the triplet and singlet states. These processes exhibit strong electron correlation, necessitating many-body approaches for an accurate description [94–96]. Such a detailed study of the spin-state readout mechanism is beyond the scope of the present paper, but some outlook can still be offered. First,

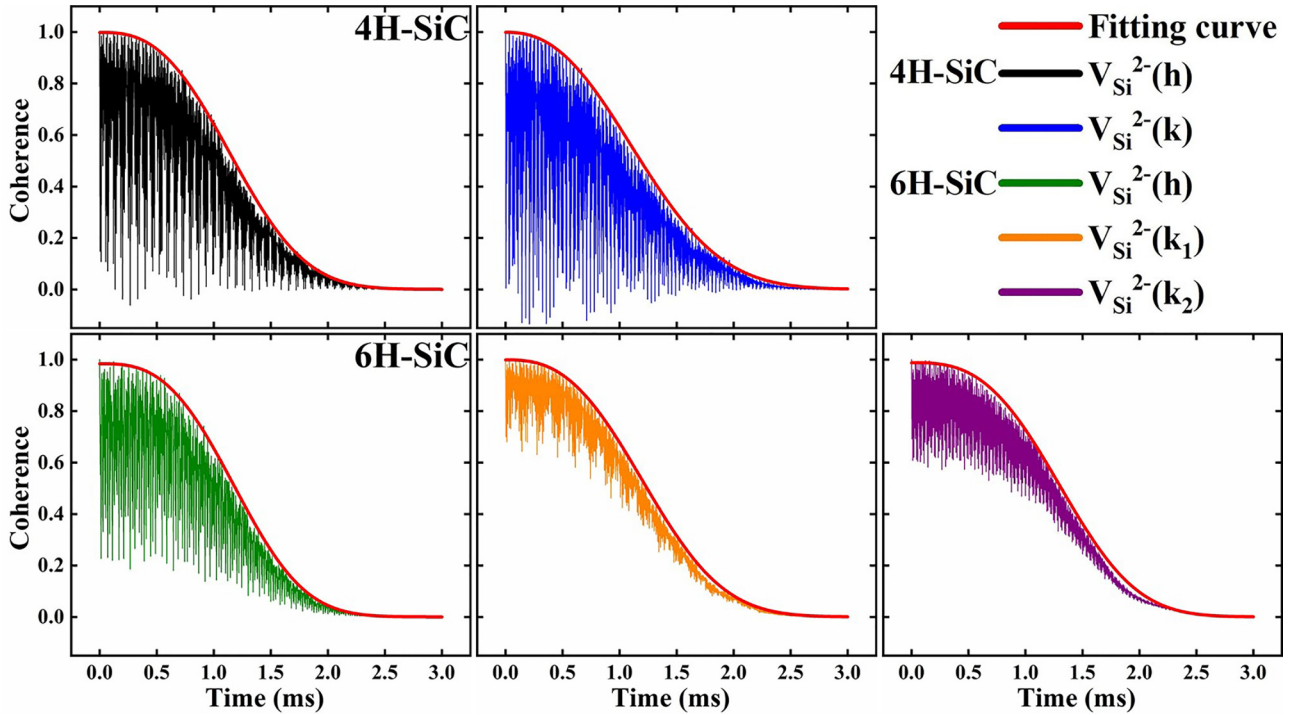


FIG. 5. Coherence function of the V_{Si}^{2-} defect in 4H- (upper panel) and 6H-SiC (lower panel). In order to extract the coherence time, this function has been exponentially fitted using equation as $\mathcal{L}(t) = e^{-(t/\tau_c)^n}$ by the red curve.

V_{Si}^{2-} in SiC exhibits a relatively short radiative decay time (55–122 ns), which is favorable for rapid optical initialization and readout. Moreover, given its NV-like spin properties, similar strategies with NV centers in SiC could potentially be employed for the initialization and readout of V_{Si}^{2-} spin states in 4H- and 6H-SiC.

B. Optical control

A key concern for the practical use of V_{Si}^{2-} in SiC as a color center is its stability and optical activity under resonant excitation. It is important to note that the stability of a given charge state is governed by different energy scales under thermodynamic equilibrium and under optical excitation. While the thermodynamic ($-2-$) charge transition level defines the equilibrium charge stability as a function of the Fermi level, optical ionization is determined by the energy separation between occupied defect levels and the CBM. In our calculations, the occupied $a_1(2)$ defect levels of V_{Si}^{2-} are located approximately 1.2–1.3 eV below the CBM in both 4H- and 6H-SiC. This indicates that direct optical ionization of the ($2-$) charge state requires photon energies well above the ZPL energies. After explicitly accounting for JT distortions, the relevant ZPLs of V_{Si}^{2-} fall in the range of 0.36–0.46 eV, providing a sufficiently broad excitation window in which resonant optical addressing can be achieved without inducing direct photoionization. Moreover, the calculated radiative lifetimes of V_{Si}^{2-} are relatively short, ranging from 55 to 122 ns, which makes it possible to observe V_{Si}^{2-} at a single defect level, indicating that the optical emission of the ($2-$) charge state is not intrinsically dim. Although off-resonant excitation at higher photon energies may enhance optical ionization

channels and reduce the effective photoluminescence, resonant or near-resonant excitation within the ZPL and its phonon sideband is expected to preserve optical activity while minimizing charge conversion. These considerations suggest that the limited experimental observation of V_{Si}^{2-} is more likely related to excitation conditions rather than an intrinsic lack of optical stability. Our results demonstrate that, when operated within an appropriate excitation window, the ($2-$) charge state can remain optically active with appreciable radiative emission, motivating further experimental studies focusing on resonant excitation and charge-state control.

In addition, it is well established experimentally that nanophotonic environments can strongly enhance the optical emission of spin defects in SiC through the Purcell effect. Previous studies have demonstrated Purcell factors approaching 50 for both VV^0 and V_{Si}^- in 4H-SiC when coupled to photonic crystal or plasmonic nanocavities, leading to significantly enhanced ZPL emission and shortened excited-state lifetimes [85,97]. These results suggest that similar cavity-based strategies could be applied to V_{Si}^{2-} to further enhance its emission rate and collection efficiency, thereby relaxing practical constraints associated with optical excitation and detection. From this perspective, the radiative properties identified here, together with established cavity-enhancement techniques in SiC, support the feasibility of optically addressing V_{Si}^{2-} as a solid-state spin-photon interface.

V. CONCLUSIONS

In conclusion, we have systematically investigated the V_{Si}^{2-} defect in 4H- and 6H-SiC by first-principles calculations. Our results show that V_{Si}^{2-} has a high-spin

ground state ($S = 1$) with electronic structure similar to NV^- and VV^0 . In terms of optical properties, V_{Si}^{2-} is an infrared color center in the IR-B and IR-C region with a large DWF ($>10\%$) due to its relatively weak electron-phonon coupling. In terms of spin properties, V_{Si}^{2-} has a large ground-state ZFS (1.16–1.29 GHz) and a long electron spin coherence time T_2 (>1.3 ms) due to its NV-like electronic structure and local environment. Taken together, we conclude that V_{Si}^{2-} combines the advantages of known color centers and can be used as a powerful qubit for quantum technology.

ACKNOWLEDGMENTS

This work was partially supported by the China National Key R&D Program (2022YFA1404603), National Natural

Science Foundation of China (Grants No. 12188101 and No. 12474222), the Guangdong Major Project of the Basic and Applied Basic Research (Future functional materials under extreme conditions 2021B0301030005), Innovation Program for Quantum Science and Technology (2024ZD0300102), and Shanghai Science and Technology Program (Grant No. 23JC1400901). Computations were performed at the High-Performance Computing Center of Fudan University.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

-
- [1] H. Ma, M. Govoni, and G. Galli, Quantum simulations of materials on near-term quantum computers, *npj Comput. Mater.* **6**, 85 (2020).
 - [2] B. Huang, N. Sheng, M. Govoni, and G. Galli, Quantum simulations of fermionic Hamiltonians with efficient encoding and ansatz schemes, *J. Chem. Theory Comput.* **19**, 1487 (2023).
 - [3] W. P. Schleich, K. S. Ranade, C. Anton, M. Arndt, M. Aspelmeyer, M. Bayer, G. Berg, T. Calarco, H. Fuchs, E. Giacobino, *et al.*, Quantum technology: From research to application, *Appl. Phys. B* **122**, 130 (2016).
 - [4] J. Wang, F. Sciarrino, A. Laing, and M. G. Thompson, Integrated photonic quantum technologies, *Nat. Photon.* **14**, 273 (2020).
 - [5] T. D. Ladd, F. Jelezko, R. Laflamme, Y. Nakamura, C. Monroe, and J. L. O'Brien, Quantum computers, *Nature (London)* **464**, 45 (2010).
 - [6] D. D. Awschalom, L. C. Bassett, A. S. Dzurak, E. L. Hu, and J. R. Petta, Quantum spintronics: Engineering and manipulating atom-like spins in semiconductors, *Science* **339**, 1174 (2013).
 - [7] G. Kurizki, P. Bertet, Y. Kubo, K. Mølmer, D. Petrosyan, P. Rabl, and J. Schmiedmayer, Quantum technologies with hybrid systems, *Proc. Natl. Acad. Sci. USA* **112**, 3866 (2015).
 - [8] G. Zhang, Y. Cheng, J.-P. Chou, and A. Gali, Material platforms for defect qubits and single-photon emitters, *Appl. Phys. Rev.* **7**, 031308 (2020).
 - [9] G. Wolfowicz, F. J. Heremans, C. P. Anderson, S. Kanai, H. Seo, A. Gali, G. Galli, and D. D. Awschalom, Quantum guidelines for solid-state spin defects, *Nat. Rev. Mater.* **6**, 906 (2021).
 - [10] Á. Gali, Recent advances in the *ab initio* theory of solid-state defect qubits, *Nanophotonics* **12**, 359 (2023).
 - [11] V. Ivády, T. Simon, J. R. Maze, I. A. Abrikosov, and A. Gali, Pressure and temperature dependence of the zero-field splitting in the ground state of NV centers in diamond: A first-principles study, *Phys. Rev. B* **90**, 235205 (2014).
 - [12] G. Balasubramanian, P. Neumann, D. Twitchen, M. Markham, R. Kolesov, N. Mizuochi, J. Isoya, J. Achard, J. Beck, J. Tissler, *et al.*, Ultralong spin coherence time in isotopically engineered diamond, *Nat. Mater.* **8**, 383 (2009).
 - [13] T. Yamamoto, T. Umeda, K. Watanabe, S. Onoda, M. L. Markham, D. J. Twitchen, B. Naydenov, L. P. McGuinness, T. Teraji, S. Koizumi, *et al.*, Extending spin coherence times of diamond qubits by high-temperature annealing, *Phys. Rev. B* **88**, 075206 (2013).
 - [14] G. Davies and M. F. Hamer, Optical studies of the 1.945 eV vibronic band in diamond, *Proc. R. Soc. London Ser. A* **348**, 285 (1976).
 - [15] Á. Gali, *Ab initio* theory of the nitrogen-vacancy center in diamond, *Nanophotonics* **8**, 1907 (2019).
 - [16] M. E. Trusheim, N. H. Wan, K. C. Chen, C. J. Ciccarino, J. Flick, R. Sundaraman, G. Malladi, E. Bersin, M. Walsh, B. Lienhard, *et al.*, Lead-related quantum emitters in diamond, *Phys. Rev. B* **99**, 075430 (2019).
 - [17] M. K. Bhaskar, D. D. Sukachev, A. Sipahigil, R. E. Evans, M. J. Burek, C. T. Nguyen, L. J. Rogers, P. Siyushev, M. H. Metsch, H. Park, *et al.*, Quantum nonlinear optics with a germanium-vacancy color center in a nanoscale diamond waveguide, *Phys. Rev. Lett.* **118**, 223603 (2017).
 - [18] B. Pingault, J. N. Becker, C. H. H. Schulte, C. Arend, C. Hepp, T. Godde, A. I. Tartakovskii, M. Markham, C. Becher, and M. Atatüre, All-optical formation of coherent dark states of silicon-vacancy spins in diamond, *Phys. Rev. Lett.* **113**, 263601 (2014).
 - [19] T. Iwasaki, Y. Miyamoto, T. Taniguchi, P. Siyushev, M. H. Metsch, F. Jelezko, and M. Hatano, Tin-vacancy quantum emitters in diamond, *Phys. Rev. Lett.* **119**, 253601 (2017).
 - [20] A. Dietrich, K. D. Jahnke, J. M. Binder, T. Teraji, J. Isoya, L. J. Rogers, and F. Jelezko, Isotopically varying spectral features of silicon-vacancy in diamond, *New J. Phys.* **16**, 113019 (2014).
 - [21] D. D. Sukachev, A. Sipahigil, C. T. Nguyen, M. K. Bhaskar, R. E. Evans, F. Jelezko, and M. D. Lukin, Silicon-vacancy spin qubit in diamond: A quantum memory exceeding 10 ms with single-shot state readout, *Phys. Rev. Lett.* **119**, 223602 (2017).
 - [22] J. N. Becker, B. Pingault, D. Groß, M. Gündoğan, N. Kukharchyk, M. Markham, A. Edmonds, M. Atatüre, P. Bushev, and C. Becher, All-optical control of the silicon-vacancy spin in diamond at millikelvin temperatures, *Phys. Rev. Lett.* **120**, 053603 (2018).

- [23] P. Siyushev, M. H. Metsch, A. Ijaz, J. M. Binder, M. K. Bhaskar, D. D. Sukachev, A. Sipahigil, R. E. Evans, C. T. Nguyen, M. D. Lukin, *et al.*, Optical and microwave control of germanium-vacancy center spins in diamond, *Phys. Rev. B* **96**, 081201(R) (2017).
- [24] K. D. Jahnke, A. Sipahigil, J. M. Binder, M. W. Doherty, M. Metsch, L. J. Rogers, N. B. Manson, M. D. Lukin, and F. Jelezko, Electron-phonon processes of the silicon-vacancy centre in diamond, *New J. Phys.* **17**, 043011 (2015).
- [25] L. J. Rogers, K. D. Jahnke, M. H. Metsch, A. Sipahigil, J. M. Binder, T. Teraji, H. Sumiya, J. Isoya, M. D. Lukin, P. Hemmer, *et al.*, All-optical initialization, readout, and coherent preparation of single silicon-vacancy spins in diamond, *Phys. Rev. Lett.* **113**, 263602 (2014).
- [26] G. Thiering and A. Gali, *Ab initio* magneto-optical spectrum of Group-IV vacancy color centers in diamond, *Phys. Rev. X* **8**, 021063 (2018).
- [27] A. Lohrmann, B. C. Johnson, J. C. McCallum, and S. Castelletto, A review on single photon sources in silicon carbide, *Rep. Prog. Phys.* **80**, 034502 (2017).
- [28] M. E. Bathan and L. Vines, Manipulating single-photon emission from point defects in diamond and silicon carbide, *Adv. Quantum Technol.* **4**, 2100003 (2021).
- [29] S. Castelletto and A. Boretti, Silicon carbide color centers for quantum applications, *J. Phys. Photon.* **2**, 022001 (2020).
- [30] S. Castelletto, L. Rosa, and B. C. Johnson, Silicon carbide for novel quantum technology devices, in *Advanced Silicon Carbide Devices and Processing*, edited by S. E. Saddow and F. La Via (InTech, London, 2015).
- [31] K. Khazen and H. J. Von Bardeleben, NV-centers in SiC: A solution for quantum computing technology?, *Front. Quantum Sci. Technol.* **2**, 1115039 (2023).
- [32] A. Cs  r  , H. J. Von Bardeleben, J.-L. Cantin, and A. Gali, Characterization and formation of NV centers in 3C, 4H, and 6H SiC: An *ab initio* study, *Phys. Rev. B* **96**, 085204 (2017).
- [33] L. Torpo, T. E. M. Staab, and R. M. Nieminen, Divacancy in 3C- and 4H-SiC: An extremely stable defect, *Phys. Rev. B* **65**, 085202 (2002).
- [34] A. Cs  r  , I. G. Ivanov, N. T. Son, and A. Gali, Fluorescence spectrum and charge state control of divacancy qubits via illumination at elevated temperatures in 4H silicon carbide, *Phys. Rev. B* **105**, 165108 (2022).
- [35] Z. Shang, A. Hashemi, Y. Berenc  n, H.-P. Komsa, P. Erhart, S. Zhou, M. Helm, A. V. Krashenninnikov, and G. V. Astakhov, Local vibrational modes of Si vacancy spin qubits in SiC, *Phys. Rev. B* **101**, 144109 (2020).
- [36] K. Sz  sz, V. Iv  dy, I. A. Abrikosov, E. Janz  n, M. Bockstedte, and A. Gali, Spin and photophysics of carbon-antisite vacancy defect in 4H silicon carbide: A potential quantum bit, *Phys. Rev. B* **91**, 121201(R) (2015).
- [37] W. F. Koehl, B. B. Buckley, F. J. Heremans, G. Calusine, and D. D. Awschalom, Room temperature coherent control of defect spin qubits in silicon carbide, *Nature (London)* **479**, 84 (2011).
- [38] Q. Li, J.-F. Wang, F.-F. Yan, J.-Y. Zhou, H.-F. Wang, H. Liu, L.-P. Guo, X. Zhou, A. Gali, Z.-H. Liu, *et al.*, Room-temperature coherent manipulation of single-spin qubits in silicon carbide with a high readout contrast, *Natl. Sci. Rev.* **9**, nwab122 (2022).
- [39] A. L. Falk, B. B. Buckley, G. Calusine, W. F. Koehl, V. V. Dobrovitski, A. Politi, C. A. Zorman, P. X.-L. Feng, and D. D. Awschalom, Polytype control of spin qubits in silicon carbide, *Nat. Commun.* **4**, 1819 (2013).
- [40] J.-F. Wang, F.-F. Yan, Q. Li, Z.-H. Liu, H. Liu, G.-P. Guo, L.-P. Guo, X. Zhou, J.-M. Cui, J. Wang, *et al.*, Coherent control of nitrogen-vacancy center spins in silicon carbide at room temperature, *Phys. Rev. Lett.* **124**, 223601 (2020).
- [41] S. A. Zargaleh, B. Eble, S. Hameau, J.-L. Cantin, L. Legrand, M. Bernard, F. Margaillan, J.-S. Lauret, J.-F. Roch, H. J. Von Bardeleben, *et al.*, Evidence for near-infrared photoluminescence of nitrogen vacancy centers in 4H-SiC, *Phys. Rev. B* **94**, 060102(R) (2016).
- [42] S. A. Zargaleh, H. J. Von Bardeleben, J. L. Cantin, U. Gerstmann, S. Hameau, B. Ebl  , and W. Gao, Electron paramagnetic resonance tagged high-resolution excitation spectroscopy of NV-centers in 4H-SiC, *Phys. Rev. B* **98**, 214113 (2018).
- [43] F. Fuchs, B. Stender, M. Trupke, D. Simin, J. Pflaum, V. Dyakonov, and G. V. Astakhov, Engineering near-infrared single-photon emitters with optically active spins in ultrapure silicon carbide, *Nat. Commun.* **6**, 7578 (2015).
- [44] D. Simin, H. Kraus, A. Sperlich, T. Ohshima, G. V. Astakhov, and V. Dyakonov, Locking of electron spin coherence over fifty milliseconds in natural silicon carbide, *Phys. Rev. B* **95**, 161201(R) (2017).
- [45] D. Simin, F. Fuchs, H. Kraus, A. Sperlich, P. G. Baranov, G. V. Astakhov, and V. Dyakonov, High-precision angle-resolved magnetometry with uniaxial quantum centers in silicon carbide, *Phys. Rev. Appl.* **4**, 014009 (2015).
- [46] M. Niethammer, M. Widmann, S.-Y. Lee, P. Stenberg, O. Kordina, T. Ohshima, N. T. Son, E. Janz  n, and J. Wrachtrup, Vector magnetometry using silicon vacancies in 4H-SiC under ambient conditions, *Phys. Rev. Appl.* **6**, 034001 (2016).
- [47] T. P. M. Alegre, C. Santori, G. Medeiros-Ribeiro, and R. G. Beausoleil, Polarization-selective excitation of nitrogen vacancy centers in diamond, *Phys. Rev. B* **76**, 165205 (2007).
- [48] L. M. Pham, N. Bar-Gill, D. Le Sage, C. Belthangady, A. Stacey, M. Markham, D. J. Twitchen, M. D. Lukin, and R. L. Walsworth, Enhanced metrology using preferential orientation of nitrogen-vacancy centers in diamond, *Phys. Rev. B* **86**, 121202(R) (2012).
- [49] S.-Y. Lee, M. Niethammer, and J. Wrachtrup, Vector magnetometry based on $S = \frac{3}{2}$ electronic spins, *Phys. Rev. B* **92**, 115201 (2015).
- [50] H. Kraus, V. A. Soltamov, F. Fuchs, D. Simin, A. Sperlich, P. G. Baranov, G. V. Astakhov, and V. Dyakonov, Magnetic field and temperature sensing with atomic-scale spin defects in silicon carbide, *Sci. Rep.* **4**, 5303 (2014).
- [51] D. Simin, V. A. Soltamov, A. V. Poshakinskiy, A. N. Anisimov, R. A. Babunts, D. O. Tolmachev, E. N. Mokhov, M. Trupke, S. A. Tarasenko, A. Sperlich, *et al.*, All-optical dc nanoscale magnetometry using silicon vacancy fine structure in isotopically purified silicon carbide, *Phys. Rev. X* **6**, 031014 (2016).
- [52] G. Kresse and J. Furthm  ller, Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set, *Phys. Rev. B* **54**, 11169 (1996).
- [53] J. Paier, M. Marsman, K. Hummer, G. Kresse, I. C. Gerber, and J. G.   ngy  n, Screened hybrid density functionals applied to solids, *J. Chem. Phys.* **124**, 154709 (2006).

- [54] J. Heyd, G. E. Scuseria, and M. Ernzerhof, Hybrid functionals based on a screened Coulomb potential, *J. Chem. Phys.* **118**, 8207 (2003).
- [55] R. Han, X. Xu, X. Hu, N. Yu, J. Wang, Y. Tian, and W. Huang, Development of bulk SiC single crystal grown by physical vapor transport method, *Opt. Mater.* **23**, 415 (2003).
- [56] T. Kimoto and J. A. Cooper, *Fundamentals of Silicon Carbide Technology: Growth, Characterization, Devices, and Applications*, 1st ed. (Wiley, New York, 2014).
- [57] A. Gali, E. Janzén, P. Deák, G. Kresse, and E. Kaxiras, Theory of spin-conserving excitation of the N–V–center in diamond, *Phys. Rev. Lett.* **103**, 186404 (2009).
- [58] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
- [59] A. Togo and I. Tanaka, First principles phonon calculations in materials science, *Scr. Mater.* **108**, 1 (2015).
- [60] M. Lax, The franck-condon principle and its application to crystals, *J. Chem. Phys.* **20**, 1752 (1952).
- [61] A. M. Stoneham, *Theory of Defects in Solids: Electronic Structure of Defects in Insulators and Semiconductors* (Oxford University Press, Oxford, 2001).
- [62] R. Kubo and Y. Toyozawa, Application of the method of generating function to radiative and nonradiative transitions of a trapped electron in a crystal, *Prog. Theor. Phys.* **13**, 160 (1955).
- [63] A. Alkauskas, B. B. Buckley, D. D. Awschalom, and C. G. Van De Walle, First-principles theory of the luminescence lineshape for the triplet transition in diamond NV centres, *New J. Phys.* **16**, 073026 (2014).
- [64] Z. Bodrog and A. Gali, The spin–spin zero-field splitting tensor in the projector-augmented-wave method, *J. Phys.: Condens. Matter* **26**, 015305 (2014).
- [65] T. Biktairov, W. G. Schmidt, and U. Gerstmann, Spin decontamination for magnetic dipolar coupling calculations: Application to high-spin molecules and solid-state spin qubits, *Phys. Rev. Res.* **2**, 022024(R) (2020).
- [66] K. Szász, T. Hornos, M. Marsman, and A. Gali, Hyperfine coupling of point defects in semiconductors by hybrid density functional calculations: The role of core spin polarization, *Phys. Rev. B* **88**, 075202 (2013).
- [67] O. V. Yazyev, I. Tavernelli, L. Helm, and U. Röthlisberger, Core spin-polarization correction in pseudopotential-based electronic structure calculations, *Phys. Rev. B* **71**, 115110 (2005).
- [68] M. Onizhuk and G. Galli, PyCCE: A python package for cluster correlation expansion simulations of spin qubit dynamics, *Adv. Theory Simul.* **4**, 2100254 (2021).
- [69] W. B. Mims, Phase memory in electron spin echoes, lattice relaxation effects in CaWO_4 : Er, Ce, Mn, *Phys. Rev.* **168**, 370 (1968).
- [70] P. A. Schultz, R. M. Van Ginhoven, and A. H. Edwards, Theoretical study of intrinsic defects in cubic silicon carbide 3C-SiC, *Phys. Rev. B* **103**, 195202 (2021).
- [71] T. Oda, Y. Zhang, and W. J. Weber, Study of intrinsic defects in 3C-SiC using first-principles calculation with a hybrid functional, *J. Chem. Phys.* **139**, 124707 (2013).
- [72] X. Yan, P. Li, L. Kang, S.-H. Wei, and B. Huang, First-principles study of electronic and diffusion properties of intrinsic defects in 4H-SiC, *J. Appl. Phys.* **127**, 085702 (2020).
- [73] E. Sörman, N. T. Son, W. M. Chen, O. Kordina, C. Hallin, and E. Janzén, Silicon vacancy related defect in 4H and 6H SiC, *Phys. Rev. B* **61**, 2613 (2000).
- [74] R. Kuate Defo, X. Zhang, D. Bracher, G. Kim, E. Hu, and E. Kaxiras, Energetics and kinetics of vacancy defects in 4H-SiC, *Phys. Rev. B* **98**, 104103 (2018).
- [75] M. E. Bathen, A. Galeckas, J. Mütting, H. M. Ayedh, U. Grossner, J. Coutinho, Y. K. Frodason, and L. Vines, Electrical charge state identification and control for the silicon vacancy in 4H-SiC, *Npj Quantum Inf.* **5**, 111 (2019).
- [76] V. Ivády, J. Davidsson, N. T. Son, T. Ohshima, I. A. Abrikosov, and A. Gali, Identification of Si-vacancy related room-temperature qubits in 4H silicon carbide, *Phys. Rev. B* **96**, 161114(R) (2017).
- [77] G. Bian, G. Thiering, and Á. Gali, Theory of optical spin-polarization of axial divacancy and nitrogen-vacancy defects in 4H-SiC, *Phys. Rev. Res.* **7**, 013320 (2025).
- [78] V. Ivády, I. A. Abrikosov, and A. Gali, First principles calculation of spin-related quantities for point defect qubit research, *npj Comput. Mater.* **4**, 76 (2018).
- [79] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/kg61-94hg> for the comparison of the PLs and spectral functions between V_{Si}^- (h) and V_{Si}^{2-} (h) in 4H-SiC, which also includes Ref. [35].
- [80] Y. Zhu, B. Kovos, M. Onizhuk, D. Awschalom, and G. Galli, Theoretical and experimental study of the nitrogen-vacancy center in 4H-SiC, *Phys. Rev. Mater.* **5**, 074602 (2021).
- [81] A. Hashemi, C. Linderålv, A. V. Krashennnikov, T. Al-Nissila, P. Erhart, and H.-P. Komsa, Photoluminescence line shapes for color centers in silicon carbide from density functional theory calculations, *Phys. Rev. B* **103**, 125203 (2021).
- [82] Y. Jin, M. Govoni, G. Wolfowicz, S. E. Sullivan, F. J. Heremans, D. D. Awschalom, and G. Galli, Photoluminescence spectra of point defects in semiconductors: Validation of first-principles calculations, *Phys. Rev. Mater.* **5**, 084603 (2021).
- [83] S. M. Schieke, P. Schroeder, and J. Krutmann, Cutaneous effects of infrared radiation: From clinical observations to molecular response mechanisms, *Photoderm. Photoimm. Photomed.* **19**, 228 (2003).
- [84] S. Castelletto and A. Boretti, Perspective on solid-state single-photon sources in the infrared for quantum technology, *Adv. Quantum Technol.* **6**, 2300145 (2023).
- [85] A. L. Crook, C. P. Anderson, K. C. Miao, A. Bourassa, H. Lee, S. L. Bayliss, D. O. Bracher, X. Zhang, H. Abe, T. Ohshima, *et al.*, Purcell enhancement of a single silicon carbide color center with coherent spin control, *Nano Lett.* **20**, 3427 (2020).
- [86] M. E. Turiansky, A. Alkauskas, M. Engel, G. Kresse, D. Wickramaratne, J.-X. Shen, C. E. Dreyer, and C. G. Van De Walle, Nonrad: Computing nonradiative capture coefficients from first principles, *Comput. Phys. Commun.* **267**, 108056 (2021).
- [87] H. J. Von Bardeleben, J. L. Cantin, E. Rauls, and U. Gerstmann, Identification and magneto-optical properties of the NV center in 4H-SiC, *Phys. Rev. B* **92**, 064104 (2015).
- [88] N. T. Son, C. P. Anderson, A. Bourassa, K. C. Miao, C. Babin, M. Widmann, M. Niethammer, J. Ul Hassan, N. Morioka, I. G. Ivanov, *et al.*, Developing silicon carbide for quantum spintronics, *Appl. Phys. Lett.* **116**, 190501 (2020).
- [89] P. Udvarhelyi, G. Thiering, N. Morioka, C. Babin, F. Kaiser, D. Lukin, T. Ohshima, J. Ul-Hassan, N. T. Son, J. Vučković, *et al.*, Vibronic states and their effect on the temperature and strain dependence of silicon-vacancy qubits in 4H-SiC, *Phys. Rev. Appl.* **13**, 054017 (2020).

- [90] T. Fazio, I. Deretzis, G. Fisicaro, E. Paladino, and A. La Magna, Stability and decoherence analysis of the silicon vacancy in 3C-SiC, *Phys. Rev. A* **109**, 022603 (2024).
- [91] H. Seo, A. L. Falk, P. V. Klimov, K. C. Miao, G. Galli, and D. D. Awschalom, Quantum decoherence dynamics of divacancy spins in silicon carbide, *Nat. Commun.* **7**, 12935 (2016).
- [92] M. Onizhuk, K. C. Miao, J. P. Blanton, H. Ma, C. P. Anderson, A. Bourassa, D. D. Awschalom, and G. Galli, Probing the coherence of solid-state qubits at avoided crossings, *PRX Quantum* **2**, 010311 (2021).
- [93] H. B. Banks, Ö. O. Soykal, R. L. Myers-Ward, D. K. Gaskill, T. L. Reinecke, and S. G. Carter, Resonant optical spin initialization and readout of single silicon vacancies in 4sH-SiC, *Phys. Rev. Appl.* **11**, 024013 (2019).
- [94] M. Bockstedte, F. Schütz, T. Garratt, V. Ivády, and A. Gali, *Ab initio* description of highly correlated states in defects for realizing quantum bits, *Npj Quantum Mater.* **3**, 31 (2018).
- [95] Y. Jin, J. Park, M. M. McMillan, D. D. Ohm, C. Barnes, B. Pingault, C. Egerstrom, B. Huang, M. Govoni, F. J. Heremans, *et al.*, First-principles framework for the prediction of inter-system crossing rates in spin defects: The role of electron correlation, *Phys. Rev. Lett.* **135**, 036401 (2025).
- [96] K. Li, V. D. Dergachev, I. D. Dergachev, S. Zhang, S. A. Varganov, and Y. Ping, Excited-state dynamics and optically detected magnetic resonance of solid-state spin defects from first principles, *Phys. Rev. B* **110**, 184302 (2024).
- [97] J.-P. So, J. Luo, J. Choi, B. McCullian, and G. D. Fuchs, Purcell enhancement and spin spectroscopy of silicon vacancy centers in silicon carbide using an ultrasmall mode-volume plasmonic cavity, *Nano Lett.* **24**, 11669 (2024).